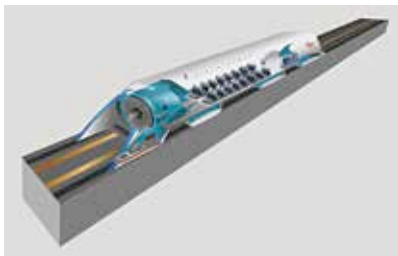


among the few countries in the world to get a functional vacetrain. Hyperloop Transportation Technologies, another Hyperloop company has also indicated its interest in building a Hyperloop in India and said that the first fully-func-



tional Hyperloop is expected in about 38 months from the time of getting greenlit by the government. Which country gets it first is purely dictated by their government.

SMART CITIES

With an ambitious plan of developing 100 cities across the country, the Union Ministry for Urban Development has undertaken on a huge responsibility of modernising our cities. Each city that gets selected under the Smart Cities Mission program has to create a corporate entity with a CEO at the helm to undertake the task of realising various projects towards the betterment of the people. The Smart Cities Mission tries to bring private sector corporate culture to state affairs. This company will receive ₹500 crore each from the State and Central government to start off and it has the option of raising additional funding from the financial market by means of debt or equity. Eighty-nine cities have so far been selected for the Smart Cities Mission program. There is no one definition for what the word “smart” stands for in this program. Instead, cities are supposed to outline goals in terms of whatever ICT and IoT infrastructure and services that they believe should be implemented in the best interests of the citizens. The Smart Cities Mission has had a few setbacks though, all cities in West Bengal, Mumbai and Navi Mumbai have withdrawn from the program for reasons political or unknown. Despite these setbacks, the Central Government remains steadfast in its goal through this program.

HAL AMCA

Set to be operational by 2030, the HAL AMCA or Advanced Medium Combat Aircraft is India's fifth-generation fighter aircraft. The Aeronautical Development Agency is leading the design aspect while HAL will be manufacturing the aircraft. The AMCA is a single-seat twin-engine stealth supermaneuverable all weather multi-role fighter. Current plans indicate that it will have a 30 mm GSh-301 cannon and have 8 or 14 hardpoints depending upon whether it's to fly in stealth configuration or not.

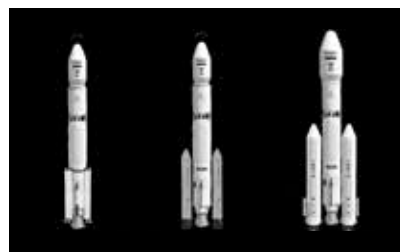


Not just the Air Force, but even the Navy has shown interest in the new Aircraft and has requested a Naval variant. The first prototype test flight was supposed to occur this year but owing to delays it'll now take place in 2025. Overall, the AMCA is estimated to cost approximately ₹9,000 crore to develop.

ULV

If you've been keeping up with ISRO's developments and read our previous story, then you might have wondered why the ULV was missing from that story. The simple reason is that it's a lot more momentous a project. The Unified Launch Vehicle is an ongoing project by ISRO which as the name suggests will be a replacement for all the different class of launch vehicles operated by ISRO. The ULV will, thus, bid goodbye to the PSLV, GSLV and LVM families of launchers.

The ULV is slated to have a common core and upper stage along with four different booster sizes. All these boosters will be solid motors. Three of these will be minor modifications of the existing boosters used by the aforementioned launch vehicles which are due to be phased out by the ULV. Additionally, there will be a heavy-lift off variant which can place payloads of up to 9 metric tonnes into Geosynchronous Trans-



fer Orbit. The ULV has so far only been mentioned a couple of times in ISRO's Annual Report so there's still a long way to go before it becomes a reality.

FAIR

This one is as sciency as it can get but it's not a wholly Indian project. Nine countries including India have come together to build the Facility for Antiproton and Ion Research (FAIR). FAIR will be an accelerator facility which will conduct experiments involving antiprotons and ions in the fields of nuclear, atomic and anti-matter physics, hadron and particle physics, high-density plasma physics and applications in condensed matter physics, and biology and biomedical sciences. The facility is under construction in Darmstadt, Germany and already has



about 3,000 scientists from more than 50 countries working on planning and designing the facilities.

THERE ARE MORE

There's no doubt that we will have missed out some pretty interesting future projects, in which case, feel free to write to us about the same. The Indian research scenario is mostly focussed on projects undertaken or supported by the government. The private sector is either getting its research done elsewhere or is really secretive about their projects, so if you're in the know about such projects then please do drop us an email and we'll surely check them out. 